

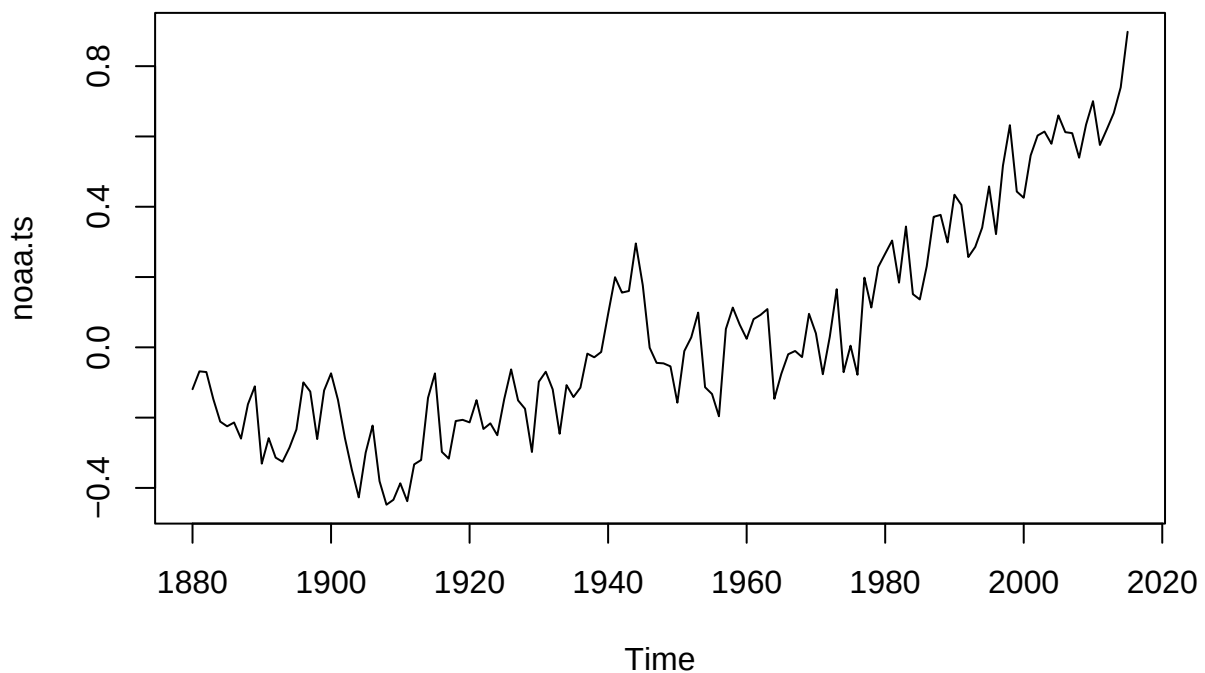
Lab: regression model for time series analysis

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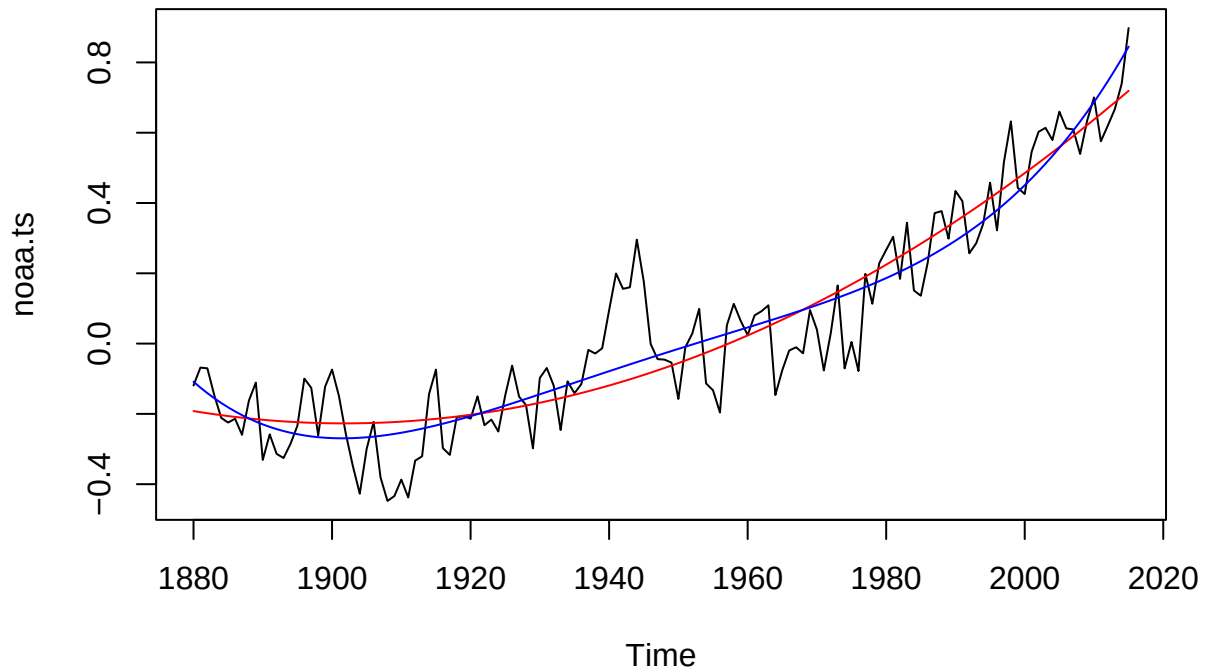
Noaa data

```
noaa.dat<-read.csv("1880-2016.csv",skip=3)
noaa.ts<-ts(noaa.dat$Value,start=1880)
plot(noaa.ts)
```



```
tt<-time(noaa.ts)
tt<-as.numeric(tt)
fit.2<-lm(noaa.ts~tt+I(tt^2))
plot(noaa.ts)
lines(tt,predict(fit.2),col="red")
fit.4<-lm(noaa.ts~poly(tt,4,raw=TRUE))
```

```
lines(tt,predict(fit.4),col="blue")
```



```
summary(fit.2)
```

```
##
## Call:
## lm(formula = noaa.ts ~ tt + I(tt^2))
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-0.25756	-0.08788	0.00596	0.06947	0.39109

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.668e+02	2.716e+01	9.823	<2e-16 ***
tt	-2.808e-01	2.790e-02	-10.065	<2e-16 ***
I(tt^2)	7.383e-05	7.163e-06	10.307	<2e-16 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1151 on 133 degrees of freedom
## Multiple R-squared:  0.8614, Adjusted R-squared:  0.8593
## F-statistic: 413.2 on 2 and 133 DF,  p-value: < 2.2e-16
```

```
summary(fit.4)
```

```
##
## Call:
## lm(formula = noaa.ts ~ poly(tt, 4, raw = TRUE))
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-0.23104	-0.07150	0.00087	0.06892	0.34832

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.300e+05  8.270e+04   3.990 0.000109 ***
## poly(tt, 4, raw = TRUE)1 -6.785e+02  1.699e+02  -3.993 0.000108 ***
## poly(tt, 4, raw = TRUE)2  5.232e-01  1.309e-01   3.996 0.000107 ***
## poly(tt, 4, raw = TRUE)3 -1.793e-04  4.482e-05  -3.999 0.000106 ***
## poly(tt, 4, raw = TRUE)4  2.304e-08  5.754e-09   4.003 0.000104 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1092 on 131 degrees of freedom
## Multiple R-squared:  0.8771, Adjusted R-squared:  0.8734
## F-statistic: 233.8 on 4 and 131 DF,  p-value: < 2.2e-16
anova(fit.2,fit.4)
```

```
## Analysis of Variance Table
##
## Model 1: noaa.ts ~ tt + I(tt^2)
## Model 2: noaa.ts ~ poly(tt, 4, raw = TRUE)
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     133 1.7634
## 2     131 1.5629  2    0.20056 8.4057 0.0003676 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
AIC(fit.2)
```

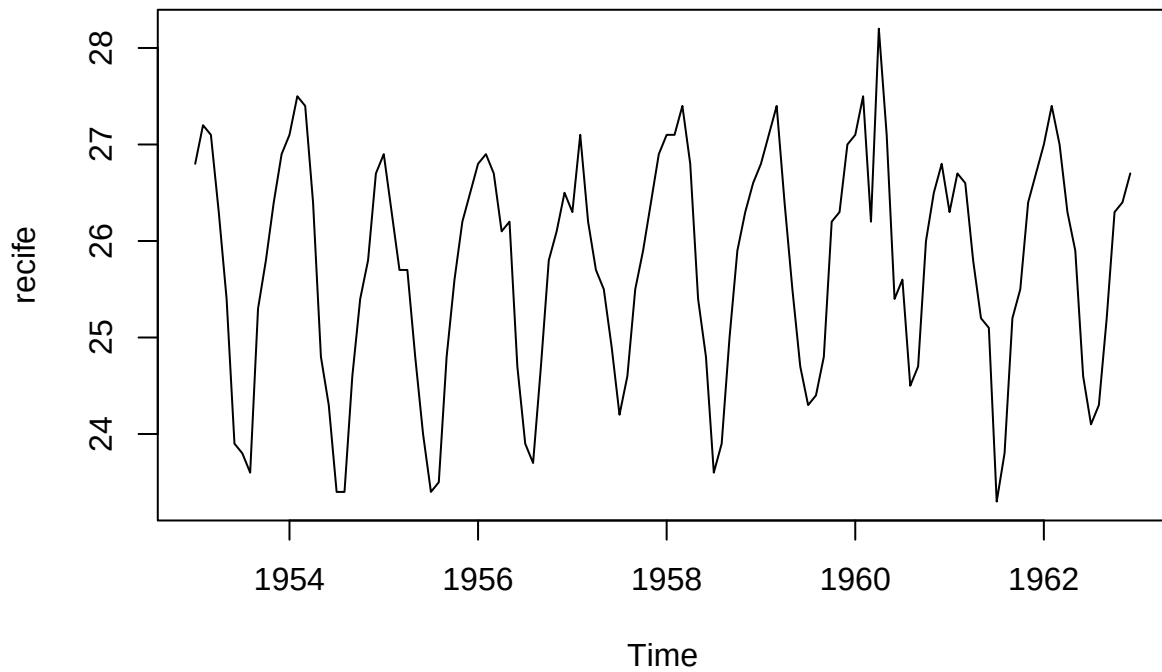
```
## [1] -197.0231
```

```
AIC(fit.4)
```

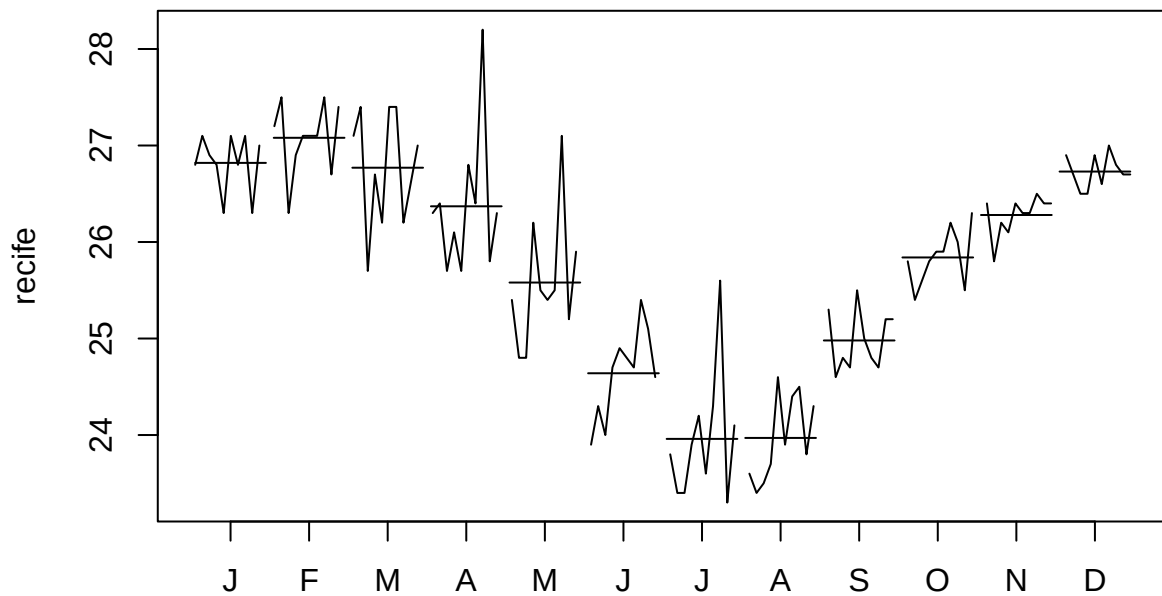
```
## [1] -209.4436
```

Recife data

```
recife<-ts(scan("recife.txt"),start = c(1953,1),frequency = 12)
plot(recife)
```



```
monthplot(recife)
```



```
library(xts)
```

```
## Loading required package: zoo
```

```
##
```

```
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## as.Date, as.Date.numeric
```

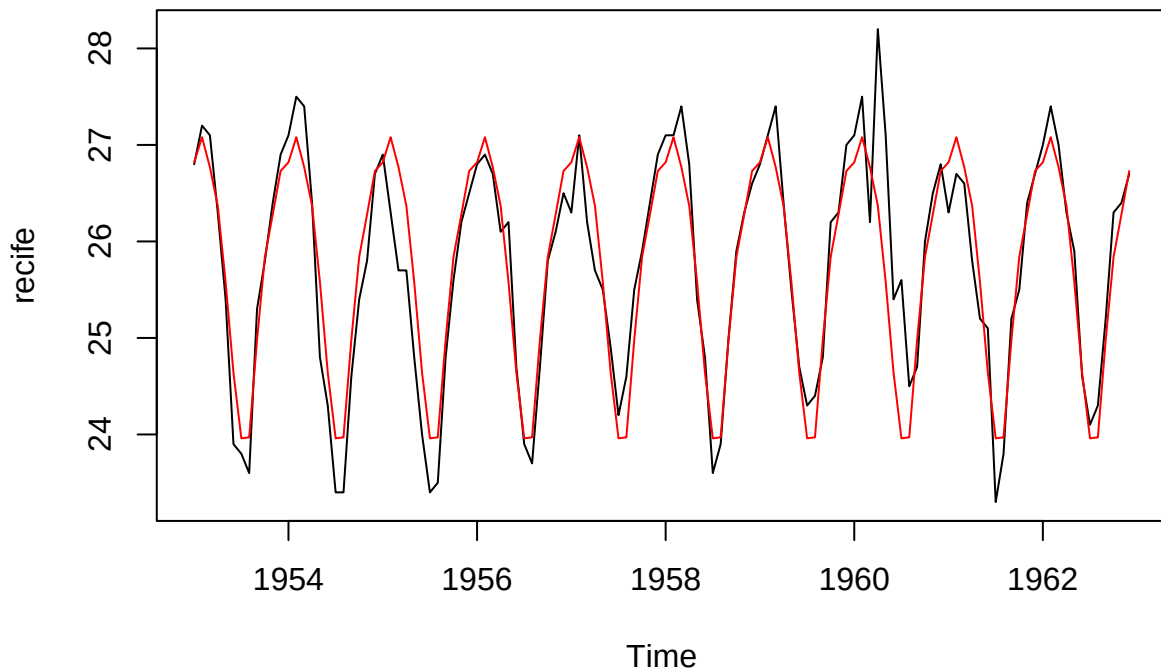
```
recife.xts<-as.xts(recife)
```

```
months<-cycle(recife.xts)
```

```
fac<-as.factor(months)
fac
```

```
##      [1] 1  2  3  4  5  6  7  8  9 10 11 12 1  2  3  4  5  6  7  8  9 10 11 12 1
##     [26] 2  3  4  5  6  7  8  9 10 11 12 1  2  3  4  5  6  7  8  9 10 11 12 1  2
##     [51] 3  4  5  6  7  8  9 10 11 12 1  2  3  4  5  6  7  8  9 10 11 12 1  2  3
##     [76] 4  5  6  7  8  9 10 11 12 1  2  3  4  5  6  7  8  9 10 11 12 1  2  3  4
##    [101] 5  6  7  8  9 10 11 12 1  2  3  4  5  6  7  8  9 10 11 12
## Levels: 1 2 3 4 5 6 7 8 9 10 11 12
```

```
fit.s<-lm(recife~fac)
plot(recife)
tt<-as.numeric(time(recife))
lines(tt,predict(fit.s),col="red")
```



Harmonic regression

We need to install the package forecast

```
install.packages("forecast")
```

```
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo

## Registered S3 methods overwritten by 'forecast':
##   method from
##   head.ts stats
##   tail.ts stats
```

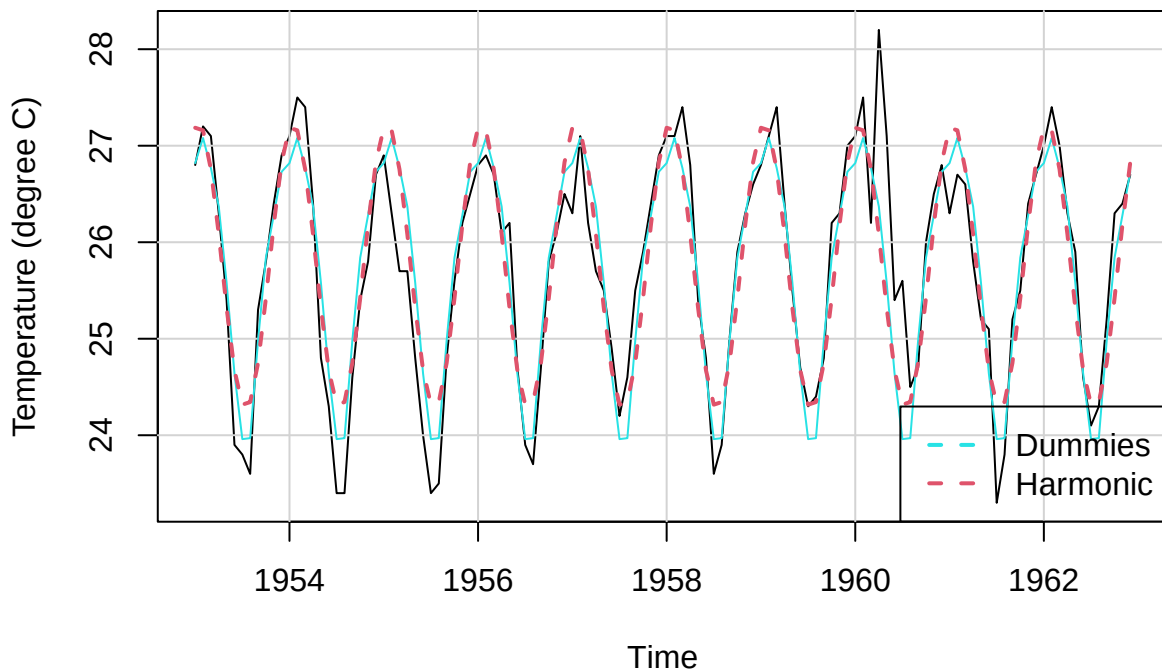
Example with one harmonic (i.e. $K = 1$, in the slides)

```

plot(recife,ylab='Temperature (degree C)',main='Recife, Brazil Temperature Data')
grid(lty=1)
K<-1
h1 <- fourier(recife, K = K)
fit.h1<-lm(recife~h1)
pred.h1<-predict(fit.h1)
lines(as.numeric(time(recife)),predict(fit.s),col=5)
lines(as.numeric(time(recife)),pred.h1,col=K+1,lwd=2,lty=2)
legend("bottomright",legend = c("Dummies","Harmonic"),lwd=2,col=c(5,2),lty=2)

```

Recife, Brazil Temperature Data



Example with two and three harmonics $K = 2, 3$

```

K<-2
h2 <- fourier(recife, K = K)
fit.h2<-lm(recife~h2)
pred.h2<-predict(fit.h2)
K<-3
h3 <- fourier(recife, K = K)
fit.h3<-lm(recife~h3)
pred.h3<-predict(fit.h3)

```

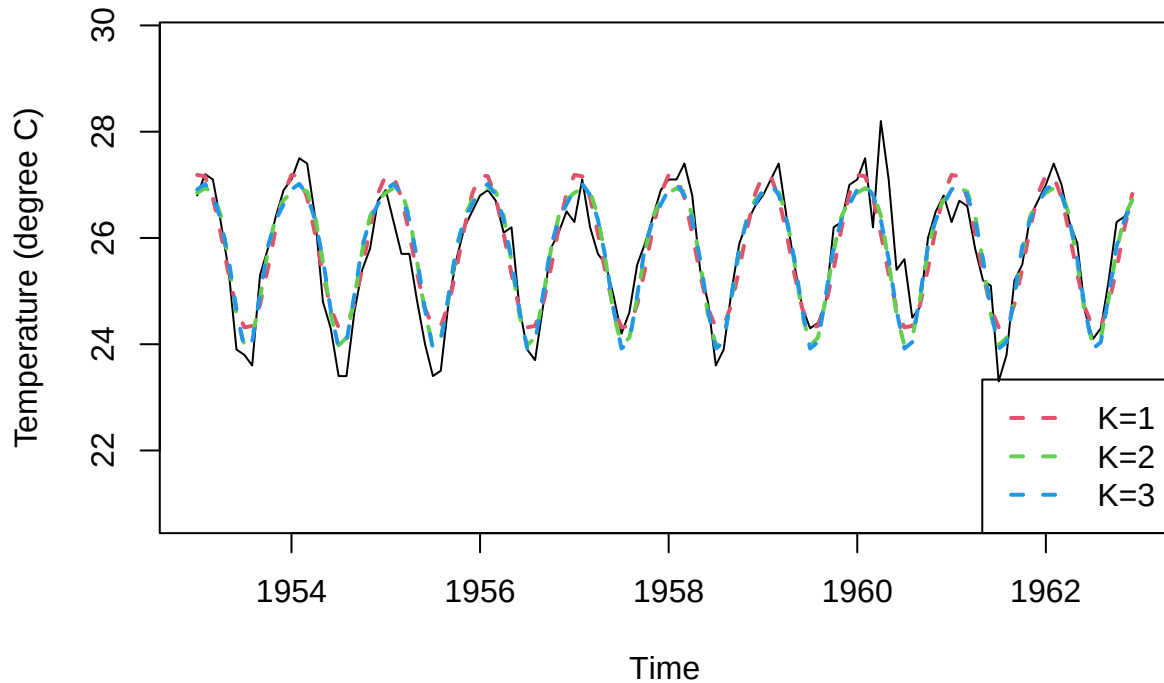
Plot of the results

```

plot(recife,ylab='Temperature (degree C)',main='Recife, Brazil Temperature Data',ylim=(range(recife)+c(
lines(as.numeric(time(recife)),pred.h1,col=2,lwd=2,lty=2)
lines(as.numeric(time(recife)),pred.h2,col=3,lwd=2,lty=2)
lines(as.numeric(time(recife)),pred.h3,col=4,lwd=2,lty=2)
legend("bottomright",legend = c("K=1","K=2","K=3"),lwd=2,col=c(2,3,4),lty=2)

```

Recife, Brazil Temperature Data



Time series decomposition: Keeling curve

See <https://www.esrl.noaa.gov/gmd/ccgg/trends> where you can download the data directly

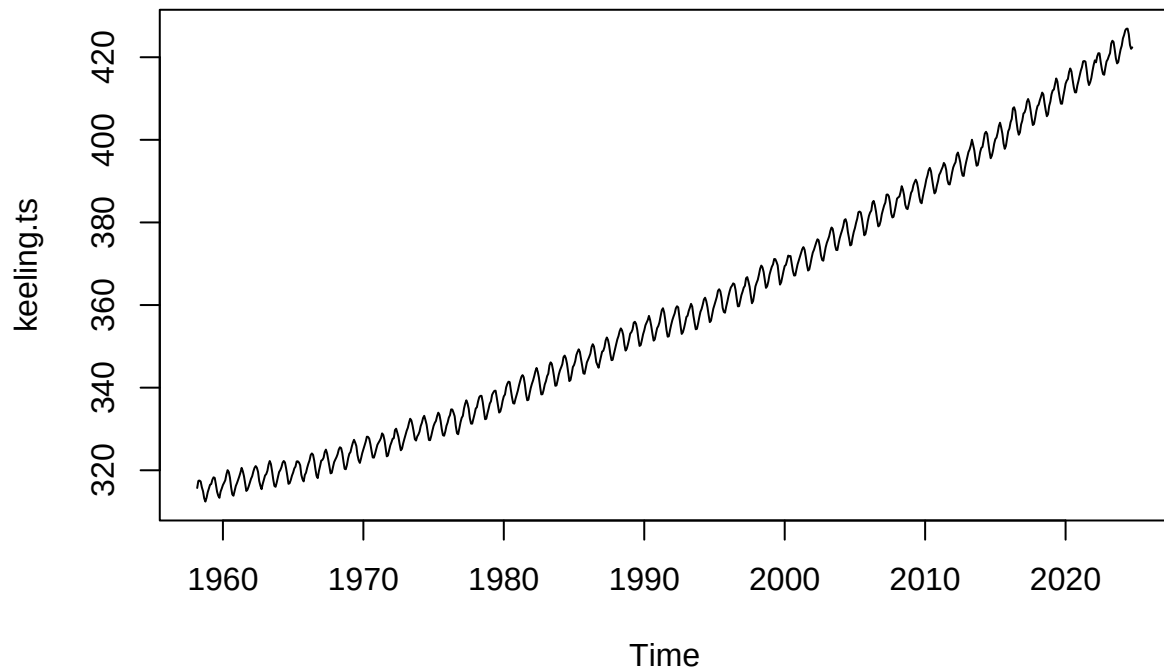
```
keeling<-read.table("https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_mm_mlo.txt",skip = 42)
head(keeling)
keeling.ts<-ts(keeling[,4],start = c(1958,3),frequency = 12)
plot(keeling.ts)
```

Alternatively

```
keeling<-read.table("co2_mm_mlo.txt",skip = 42)
head(keeling)
```

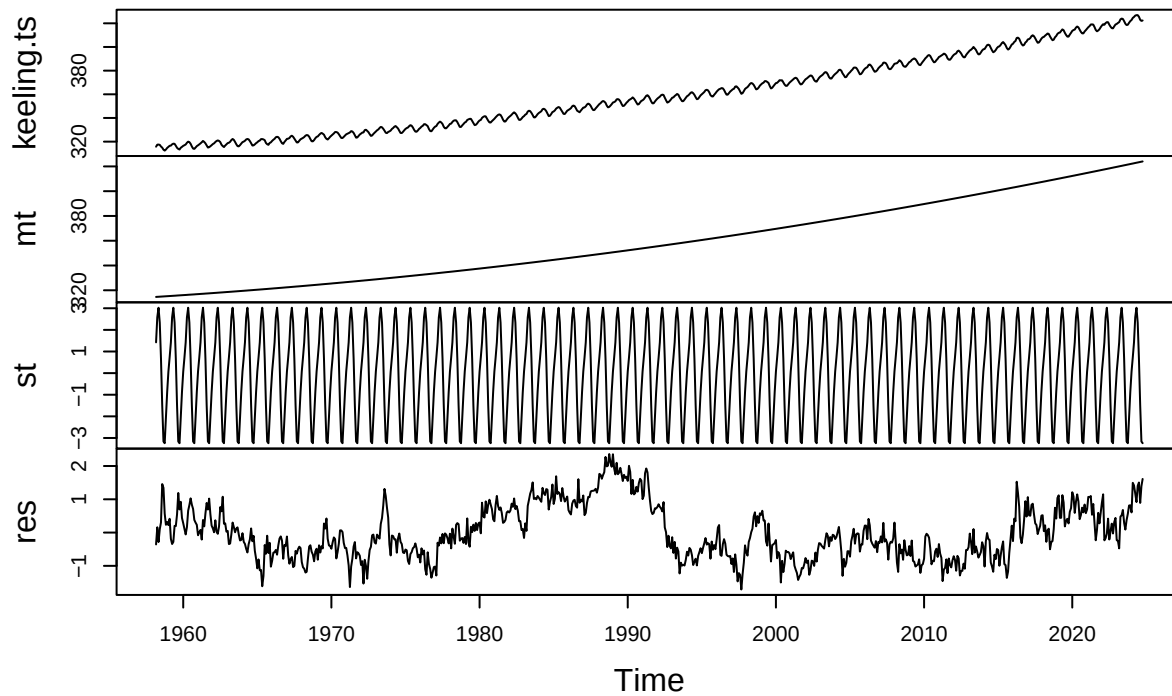
```
##      V1 V2      V3      V4      V5 V6      V7      V8
## 1 1958  3 1958.203 315.71 314.44 -1 -9.99 -0.99
## 2 1958  4 1958.288 317.45 315.16 -1 -9.99 -0.99
## 3 1958  5 1958.370 317.51 314.69 -1 -9.99 -0.99
## 4 1958  6 1958.455 317.27 315.15 -1 -9.99 -0.99
## 5 1958  7 1958.537 315.87 315.20 -1 -9.99 -0.99
## 6 1958  8 1958.622 314.93 316.21 -1 -9.99 -0.99
```

```
keeling.ts<-ts(keeling[,4],start = c(1958,3),frequency = 12)
plot(keeling.ts)
```



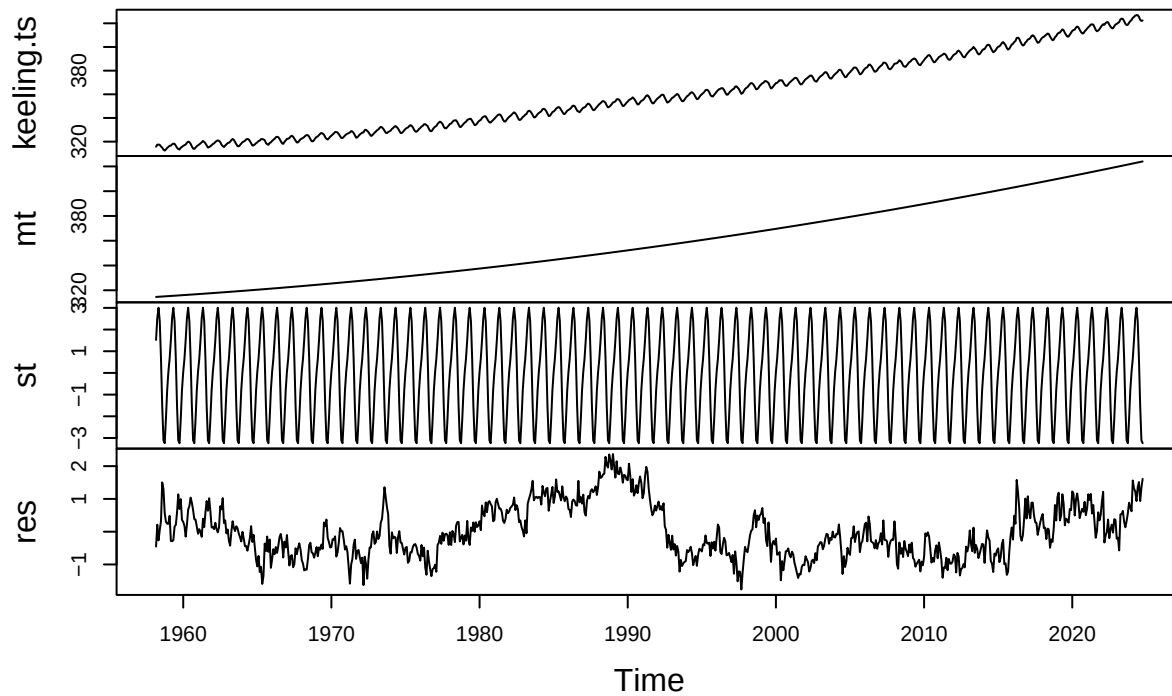
```
keeling.xts<-as.xts(keeling.ts)
months<-cycle(keeling.ts)
fac<-as.factor(months)
tt<-as.numeric(time(keeling.ts))
tt2<-tt^2
fit.s<-lm(keeling.ts~tt+tt2+C(fac,sum))
beta<-coef(fit.s)
mt<-beta[1]+beta[2]*tt+beta[3]*tt2
res<-resid(fit.s)
st<-keeling.ts-mt-res
d<-ts(cbind(keeling.ts,mt,st,res),start = start(keeling.xts),frequency = frequency(keeling.xts))
plot(d,main="Decomposition")
```


Decomposition



```
K<-3
h<-fourier(keeling.ts, K = K)
fit.h<-lm(keeling.ts~tt+tt2+h)
beta<-coef(fit.h)
mt<-beta[1]+beta[2]*tt+beta[3]*tt2
res<-resid(fit.h)
st<-keeling.ts-mt-res
d<-ts(cbind(keeling.ts,mt,st,res),start = start(keeling.xls),frequency = frequency(keeling.xls))
plot(d,main="Decomposition with harmonics")
```

Decomposition with harmonics



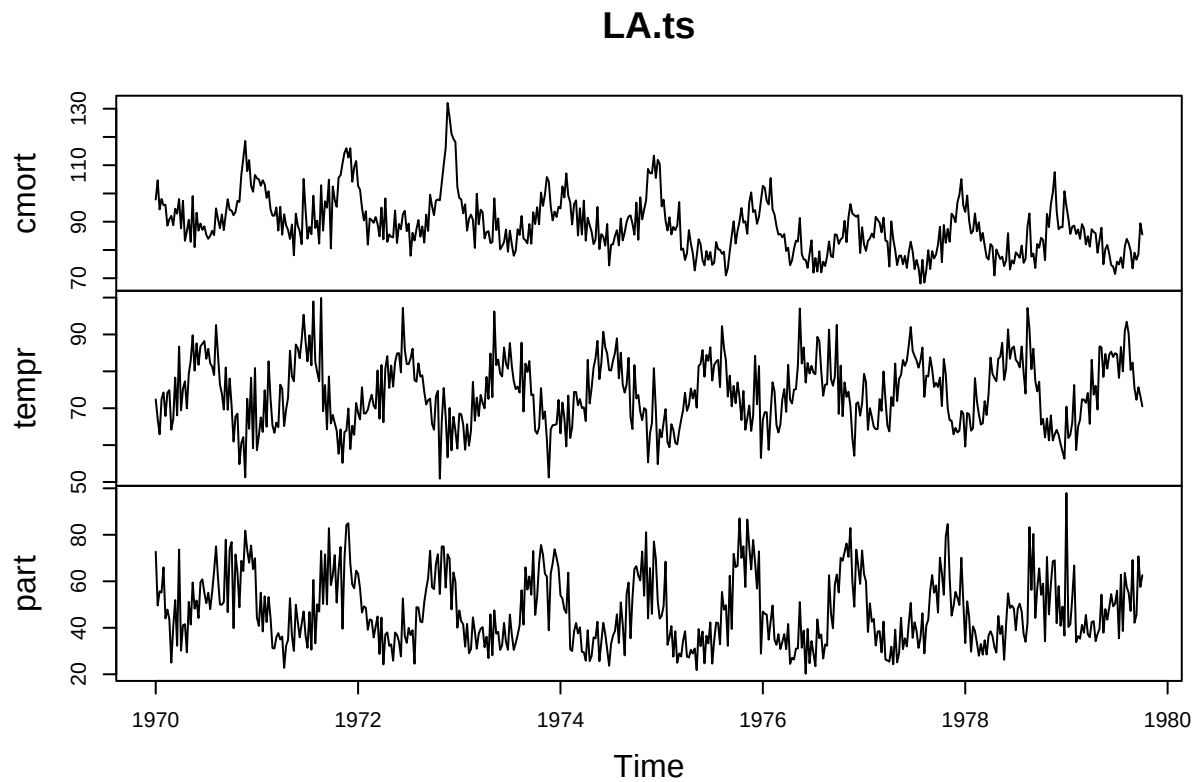
An epidemiological example

We study the possible effects of temperature and pollution on weekly mortality in Los Angeles County.

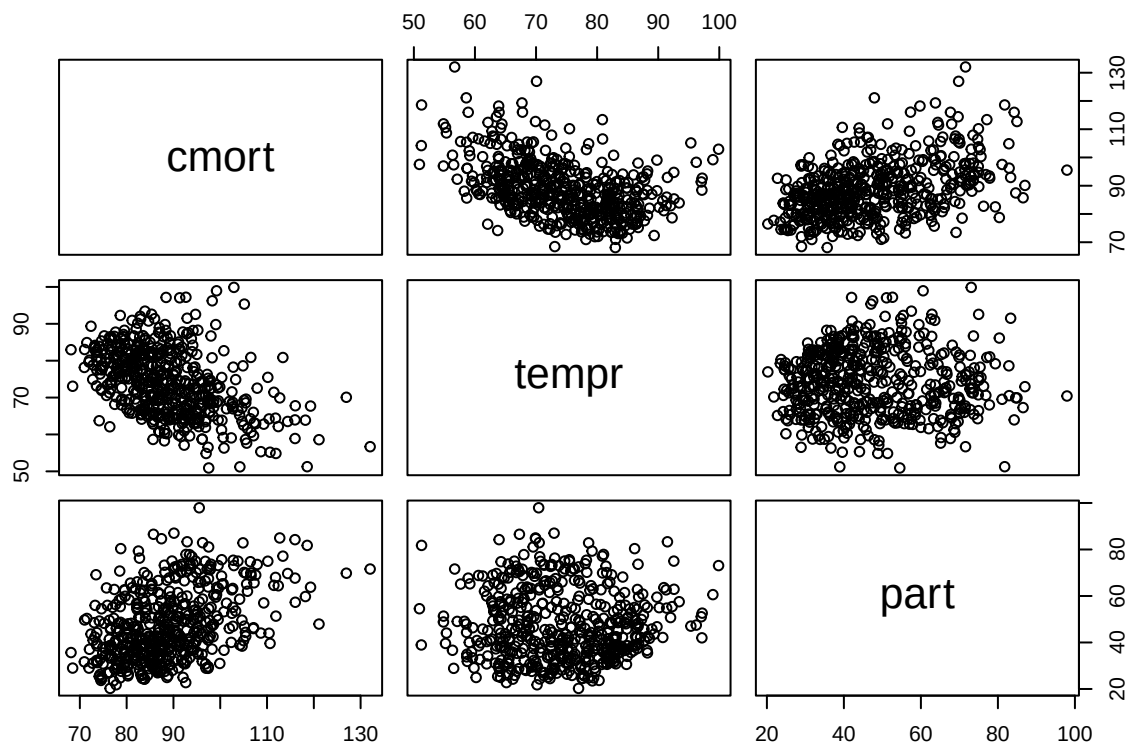
```
LA<-read.csv("LA.csv")
```

```
LA.ts<-ts(LA,start =c(1970,1),end=c(1979,40),frequency = 52)
```

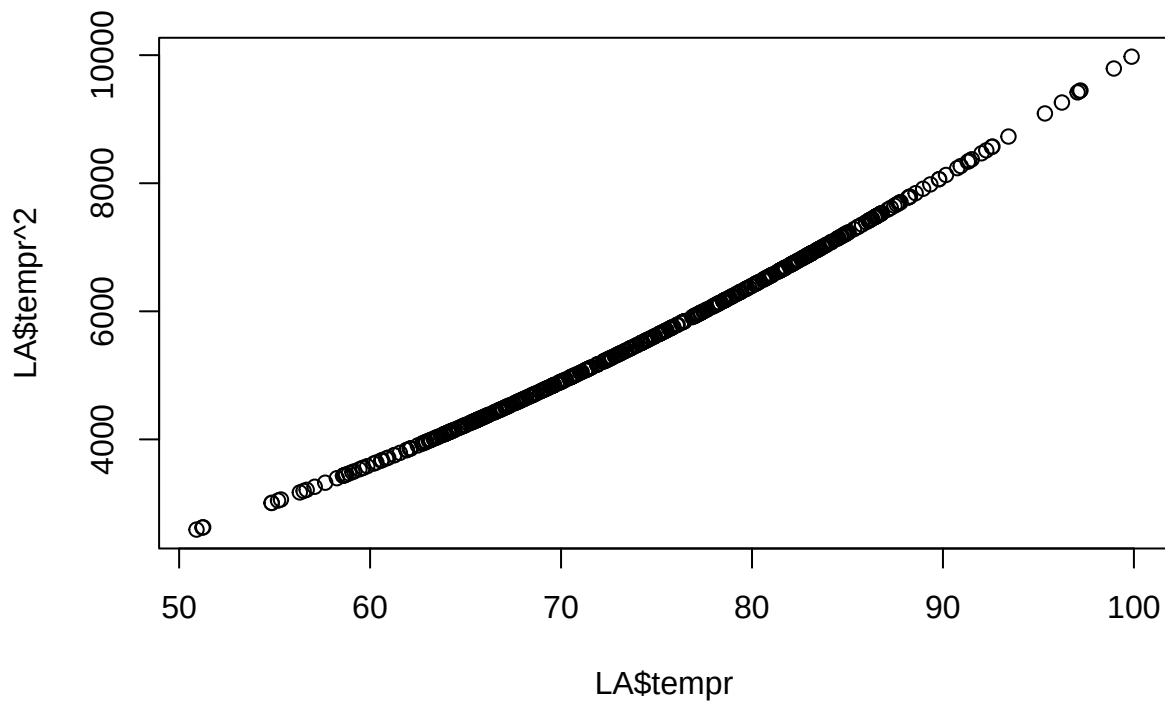
```
plot(LA.ts)
```



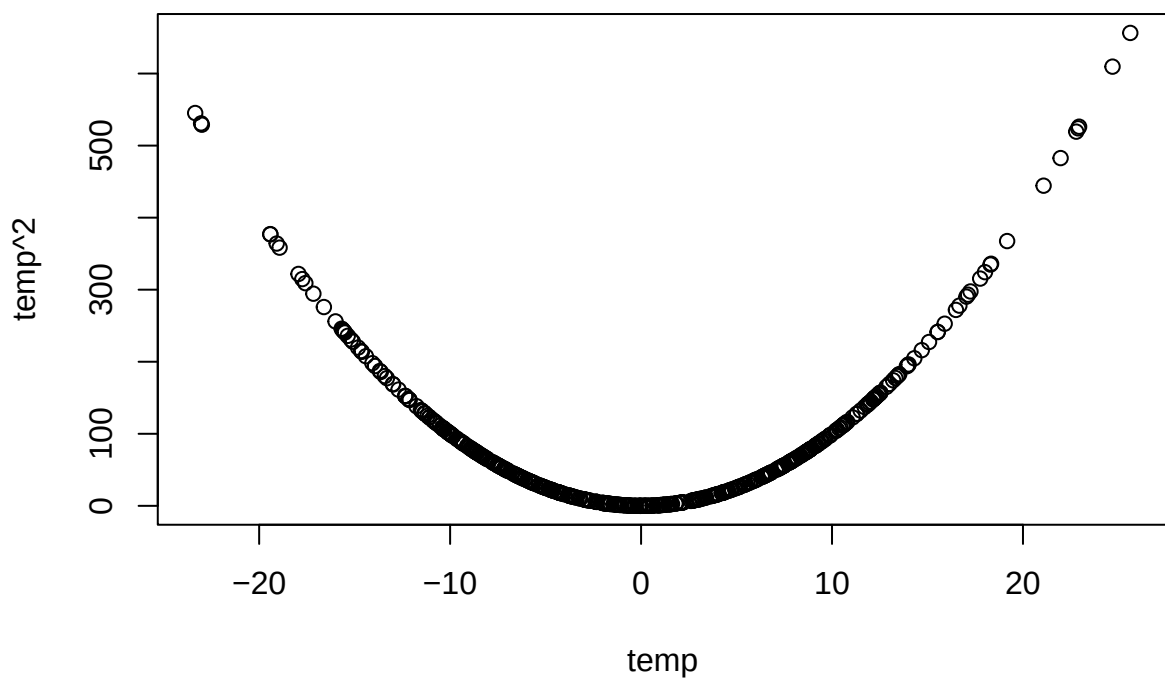
```
pairs(LA)
```



```
plot(LA$tempr, LA$tempr^2) # collinear
```



```
a<-cor(LA$tempr, LA$tempr^2)
temp <- LA$tempr - mean(LA$tempr)
plot(temp, temp^2) # not collinear
```



```
b<-cor(temp, temp^2)

temp2<-temp^2
trend <- time(LA.ts) # time is trend
fit <- lm(cmort~ trend + temp + temp2 + part,data = LA)
summary(fit) # regression results
```

```
##
## Call:
## lm(formula = cmort ~ trend + temp + temp2 + part, data = LA)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-19.0760	-4.2153	-0.4878	3.7435	29.2448

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.831e+03	1.996e+02	14.19	< 2e-16 ***
trend	-1.396e+00	1.010e-01	-13.82	< 2e-16 ***
temp	-4.725e-01	3.162e-02	-14.94	< 2e-16 ***
temp2	2.259e-02	2.827e-03	7.99	9.26e-15 ***
part	2.554e-01	1.886e-02	13.54	< 2e-16 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.385 on 503 degrees of freedom
## Multiple R-squared:  0.5954, Adjusted R-squared:  0.5922
## F-statistic: 185 on 4 and 503 DF, p-value: < 2.2e-16
```